Context-Preserving Computational Assessment of Redox Motifs for Polymer Grafting

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Abstract

Selecting redox motifs for post-functionalization of a polymer requires molecular electrochemistry, attachment chemistry and neighboring-site behavior to be evaluated for compatible structures and conditions. We present a context-preserving computational framework that links exact precursor and graft identities to state-specific free energies, balanced attachment reactions and finite oligomer occupancy cycles, with numerical qualification distinguished from experimental accuracy. A retrospective benchmark of 39 acetonitrile potentials gives a mean absolute error of 0.247 V; an exact viologen graft has calculated reductions at −0.608 and −1.208 V versus ferrocene and a separated-ion local attachment free energy of −11.49 kJ mol^{-1}, while a PMDI case shows a 5.3–6.1 kJ mol^{-1} first-site polymer penalty and −1.86 kJ mol^{-1} neighboring-site nonadditivity within sampled minima. Together with asymmetric TEMPO solution-relaxation costs and close-contact phthalimide conformers at two sequence spacings, these results identify which molecular, reaction and polymer comparisons are supported, and which remain conditional on speciation, sampling and material accessibility.

Keywords: molecular modeling; redox-active polymers; chemical identity; grafting thermodynamics; density functional theory; conformer sampling; computational electrochemistry; reproducibility.

1 Introduction

Redox-active polymers offer a means of retaining charge-storage groups within macromolecular or particulate architectures while allowing their electrochemistry to be tuned through molecular design. Aqueous polymer flow batteries, nonaqueous redox-active colloids and insoluble beads coupled to soluble mediators illustrate different implementations of this idea.[1–3] Post-functionalization of an existing support is particularly attractive when the polymer architecture can be prepared independently of the redox group. Phenothiazine- and TEMPO-functionalized materials derived from chloromethylated commodity polystyrene further demonstrate the practical relevance of this route.[4] Selecting a suitable pendant group nevertheless requires more than a favorable potential of the free parent molecule.

Three changes accompany attachment to a preformed support. The redox core becomes an exact grafted molecular structure with a particular bond and linker; its installation requires an accessible precursor and defined reaction conditions; and its local environment depends on neighboring grafts, residual reactive sites and polymer conformation. These changes introduce questions that are often evaluated in different models. A molecular reduction potential, a base-assisted substitution energy and an oligomer interaction energy may each be well defined while remaining unsuitable for direct combination. Treating all three as interchangeable indicators of a promising polymer can obscure the assumptions governing the prediction.

Existing computational studies provide important parts of this assessment. High-throughput quinone screening and redox databases organize chemical space and connect structure to electrochemical descriptors.[5–7] Achazi and co-workers developed multistep screening of redox-active molecules for radical-polymer anodes and flow-battery anolytes, combining conformer sampling, electronic-structure filters and benchmarked potentials.[8] Computational electrochemistry also establishes the importance of reference conventions, solvation and charge-state treatment.[9–11] At the polymer level, association of diimides, electron hopping, and concentration- and backbone-dependent transport show why neighboring groups and chain structure matter.[12–14] General provenance systems make calculations traceable and reusable, but provenance alone does not determine whether two chemical quantities have compatible meanings.[15–17]

Here we connect these levels through an assessment in which each property retains its exact chemical identity and calculation context. Molecular states are linked to the precursor/product pair and new bond of a grafting reaction, while balanced differences compare the same graft across a capped site and explicit open oligomers. The approach distinguishes a common base contribution, a first-site polymer-context change and a neighboring-site response. These are applications of established thermodynamic cycles; the methodological contribution is their coordinated use with explicit identity, model and quality requirements, so that a supported comparison can be distinguished from an untested transfer of a descriptor.

We evaluate this construction through a retrospective redox benchmark and complementary candidate-level examples. Viologen links a demonstrated attachment class to matched molecular and local-reaction calculations. TEMPO tests the interpretation of directional solution relaxation. PMDI connects an enumerated molecular design to first-site and neighboring-site polymer cycles, and phthalimide controls examine whether a sequence spacer prevents spatial contact. The study asks what these linked calculations establish about a proposed graft, rather than assuming that every favorable descriptor constitutes material validation.

2 Computational Framework and Methods

2.1 Chemical identities and permissible property combinations

The target synthesis is post-functionalization of a preformed polymer bearing benzyl chloride handles. The local model is 4-methylbenzyl chloride, whose reaction with a motif gives the corresponding capped graft. This model specifies local connectivity without representing a complete bead, crosslink network or graft distribution. Free parents, capped grafts and oligomers are distinct chemical objects throughout the analysis.

A calculated property is associated with a structured record,

$$\mathcal{P} = \{x, \mathcal{I}, \mathcal{C}, \mathcal{Q}, \mathcal{S}\}, \tag{1}$$

where x is the value, $\mathcal{I}$ identifies the molecular graph, charge, multiplicity and geometry, $\mathcal{C}$ defines the electronic and thermal methods, solvent, temperature, standard state and reference, $\mathcal{Q}$ records the applicable quality checks, and $\mathcal{S}$ binds the source records. A property can be reused for the same exact structure only when its state and model requirements match the new calculation. Reference subtraction and reaction sums impose additional compatibility conditions. A failed or absent check leaves the corresponding quantity unresolved; no missing thermal term is replaced by zero.

Candidate generation uses RDKit 2026.3.5 and exact stereochemical graphs.[18] Every generation recipe is retained after deduplication. Attachment is defined by a precursor/product pair and a mapped new heavy-atom bond. The route assessment requires evidence for exact precursor

access, attachment selectivity, reaction conditions, a balanced route and redox-core retention. An exact-support demonstration, a molecular analogue and a proposed transformation retain different evidential scopes. Route admission permits a stated next calculation stage; it does not establish conversion or material performance.

The frozen systematic cohort comprises viologen, pyromellitic-diimide (PMDI) and naphthalene-diimide (NDI) cores with declared linkers, terminal groups and single ring substitutions. Later proposal panels and evaluated structures are joined by exact graft graph while retaining aliases and source roles. The 904-graph maintained registry is an identity inventory, not 904 completed property evaluations. Nominal molecular capacity is calculated from $Q = nF/(3.6M)$, where M is molar mass in g mol^{-1} and n is the assumed electron count. Accessible electron count and functionalized-polymer mass require separate evidence.

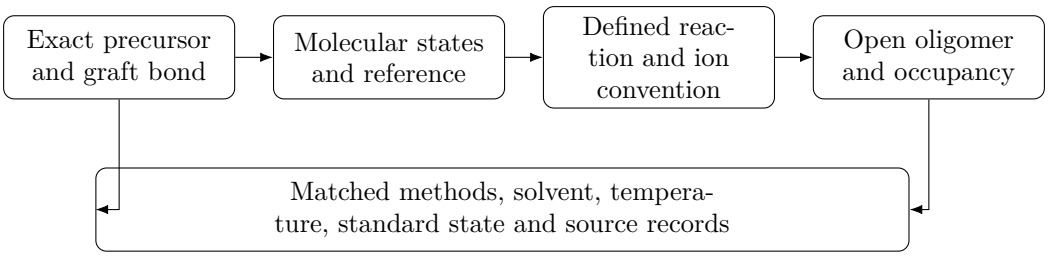


Figure 1: Context-preserving assessment of a graft. Exact structures connect molecular electrochemistry, attachment reactions and conditional oligomer comparisons. Each value retains its own state and calculation context; compatibility and physical quality checks govern permitted combinations. A passed route or molecular calculation does not supply missing polymer or material evidence.

2.2 Electronic structure, thermochemistry and physical acceptance

Geometry optimization and native Hessians use r^{2}SCAN-D4 with SMD acetonitrile, employing def2-SVP for neutral/cationic species and def2-SVPD for anions.[19–21] The final electronic energies use a uniform ωB97M-V/def2-TZVPD protocol for all molecular charge states and the ferrocene reference.[22–24] Calculations use PySCF 2.14.0, GPU4PySCF 1.8.1 and geomeTRIC.[25–27] Density fitting uses the implementation-default auxiliary bases with `only_dfj=False`, fitting both Coulomb and exact-exchange contributions in hybrid calculations. Supporting electrolyte ions are absent unless an explicit ionic model is specified.

At 298.15 K the composite free energy is

$$G = E_{\mathrm{SMD}}^{\omega\mathrm{B97M-V}} + G_{\mathrm{nuclear}}^{\mathrm{qRRHO}} - RT\ln(2S+1). \tag{2}$$

The nuclear term is obtained from the geometry-level solution DFT Hessian. Mass-weighted projection removes rigid translations and rotations, and the xTB 6.7.1 thermodynamic kernel integrates the explicit internal-mode list using unscaled frequencies.[28, 29] Its use as an integrator does not introduce an xTB Hessian. The symmetry number is determined from the exact geometry. A 50 cm^{-1} rotor cutoff is primary, with 25 and 100 cm^{-1} settings used for sensitivity. Modes from -5 to 0 cm^{-1} enter as absolute frequencies within the stated minimum tolerance; lower frequencies fail. The spin contribution assumes an unsplit ground multiplet.

Admission requires a maximum Cartesian gradient component of 4.5×10^{-4} hartree bohr^{-1} or less, Hessian asymmetry of 10^{-6} or less in the stored atomic-unit matrix, and a lowest projected

internal frequency no lower than -5 cm^{-1}. Two complete sampled Hessian columns are compared with displaced gradients at 0.0005 and 0.001 bohr and must satisfy a 1% relative-norm criterion; the coordinate-selection policy is given in the SI. This samples derivatives rather than validating every matrix column. Finite data, full grid response and $|\langle S^2\rangle - S(S+1)| \leq 0.1$ are checked independently. The active energy must refer to the actual accepted geometry. Incomplete or repaired structures remain unqualified until their required checks pass.

The charge-dependent geometry basis and the composite electronic/Hessian protocol are deliberate approximations. Five neutral benchmark partners were separately reoptimized and evaluated with def2-SVPD Hessians to measure one aspect of this imbalance. Their anion and ferrocene records were held fixed. Broader basis convergence, alternative spin-state thermochemistry and global conformer convergence are not inferred from this sensitivity test.

2.3 Redox potentials and retrospective validation

For a one-electron reduction, $\mathrm{O} + \mathrm{e}^- \rightarrow \mathrm{R}$, the potential relative to the computed $\mathrm{Fc/Fc^+}$ couple is

$$E_{\mathrm{O/R}}^{\mathrm{vs\ Fc}} = \frac{G_{\mathrm{O}} - G_{\mathrm{R}} - [G_{\mathrm{Fc^+}} - G_{\mathrm{Fc}}]}{F}. \tag{3}$$

Equation 3 is written for molar energies; numerical energy differences in eV per molecule give the corresponding one-electron voltage directly. Both reference states and both molecular states use matched electronic and thermal conventions. Stored reference constants serve as diagnostics rather than replacements for a missing computed reference.

The retrospective panel contains 39 sourced MeCN endpoints from 30 exact free-parent molecules and seven studies. Source location, reference conversion and evidence tier are retained for each endpoint. This is a benchmark of the chosen protocol on the available panel, not a prospective held-out test. Statistics use unrounded values without outlier removal or a fitted primary offset. Molecule- and study-cluster bootstrap intervals use 2,000 replicates with seed 20261021 and at least five clusters. Post hoc grouped offset diagnostics exclude every endpoint belonging to the held-out molecule or study. Their purpose is to expose residual structure; they do not establish a calibrated error distribution for new grafts.

2.4 Attachment reactions and matched thermodynamic differences

Neutral N–H/O–H substitution and pyridyl quaternization require different reaction definitions. For the former, the bookkeeping reference and two base-assisted models are

$$\mathrm{RCl} + \mathrm{HNu} \rightarrow \mathrm{RNu} + \mathrm{HCl}, \tag{4}$$

$$\mathrm{RCl} + \mathrm{HNu} + \mathrm{B} \rightarrow \mathrm{RNu} + \mathrm{BH^+} + \mathrm{Cl^-}, \tag{5}$$

$$\mathrm{RCl} + \mathrm{HNu} + \mathrm{B} \rightarrow \mathrm{RNu} + [\mathrm{BH^+} \cdots \mathrm{Cl^-}]. \tag{6}$$

The first equation omits a synthesis base. The other two specify separated ions or a contact pair; they do not prescribe actual populations or salt precipitation. Quaternization instead combines a pyridyl precursor with RCl to form the quaternized graft and chloride. For the mono-N-methyl bipyridinium precursor, the total charges are $(+1) + 0 \rightarrow (+2) + (-1)$, with no HCl or base-capture term.

All sums conserve atoms and charge. The nuclear translation convention is 1 atm; conversion to a 1 M solute standard state adds $\Delta n RT \ln(RTc^\circ/p^\circ)$. It cancels for the neutral reference, separated-ion substitution and separated-ion quaternization. The contact-pair substitution has $\Delta n = -1$,

giving approximately $-7.93\ \mathrm{kJ\,mol^{-1}}$ at 298.15 K. The base pilot computes neutral, protonated and chloride-paired DBU/DIPEA components from three geometry seeds, with thermochemistry at the selected electronic minimum. The viologen precursor campaign instead checks all three fixed seeds and selects the lowest precursor G at the primary cutoff, retaining every seed and cutoff result.

For matched N–H/O–H reactions a fixed base representation contributes a common capture term,

$$\Delta G_{\mathrm{graft,B}} = \Delta G_{\mathrm{graft,HCl}} + \Delta G_{\mathrm{capture,B}}. \tag{7}$$

External proton-transfer controls use graph-matched MeCN basicity/acid tables and separately reviewed DIPEA measurements.[30–34] The conversion $\Delta G^\circ = -RT\ln(10)\Delta \mathrm{p}K_a$ tests the specified component reaction. Since HCl cancels from the complete DFT substitution cycle, replacing only its capture term changes error cancellation and is interpreted as a sensitivity, not a correction to the full reaction.

For an empty open oligomer P_0, a single graft P_i, local site RCl and capped graft RM, the polymer-minus-local first-attachment difference is

$$\delta G_{i,\mathrm{poly-local}} = G(P_i) + G(\mathrm{RCl}) - G(P_0) - G(\mathrm{RM}). \tag{8}$$

Common precursor, base and byproduct terms cancel. A second difference uses the empty, singly occupied and doubly occupied graphs,

$$\Delta\Delta G_{ij} = G(P_{ij}) + G(P_0) - G(P_i) - G(P_j). \tag{9}$$

It measures the change in attachment at j when i is occupied under matched reaction conventions. Both differences include backbone, conformational and solvent response; neither isolates motif binding. Reference cancellation requires the same chemical and physical conventions, not merely energies with similar file labels.

2.5 Oligomer sampling and complementary interaction descriptors

Five-repeat spacing models contain three reactive sites and two plain-styrene padding units. Occupied sites carry the specified motif; unoccupied reactive sites retain benzyl chloride. The open backbone, graft connectivity and declared configuration are checked in both molecular graphs and actual XYZ structures. A reactive-handle density is not a graft conversion, and a deterministic occupancy scenario is not a probability distribution for a real polymer.

Finite conformer sets use the same graph-specific partition at 298.15 K,

$$G_{\mathrm{set}} = -RT\ln\sum_k \exp[-G_k/(RT)]. \tag{10}$$

Duplicate review and component-wise sampling are required before interpreting convergence of a balanced cycle. The PMDI primary cycle uses one minimum per graph; the phthalimide spacing control uses two poses per graph. Expanded sampling under a separately identified smooth-surface MeCN model supplies a diagnostic of sampling requirements. Its free energies cannot enter the primary-model cycles.

Two further descriptors address different aspects of a graft. Frozen methyl-capped phthalimide interactions use $E_{AB}-E_A-E_B$ at unchanged fragment coordinates and the same medium; these are electronic interactions without thermal association or counterpoise correction. Directional solution relaxation of the exact TEMPO graft uses a Pekar partition of SMD electrostatics, with fast

solvent polarization relaxed and the initial slow polarization frozen.[35] All six equilibrium/vertical energies share the active level. These electronic relaxation costs contain the geometry and solvent response once; no gas inner-sphere, Born outer-sphere or thermal term is added. The SI specifies dielectric constants, energy cycles and the 5% directional-asymmetry diagnostic for a symmetric approximation.

3 Results and Discussion

3.1 A linked design space has uneven validation coverage

The systematic construction gives 1,593 recipes and 837 exact graft graphs: 459 viologens, 243 NDI derivatives and 135 PMDI derivatives. Its linkers, caps and ring substitutions define a focused chemical space. The historical evaluated cohort contains 75 candidates and one comparison control; its union with enumeration comprises 879 graphs. Later registered models and proposal panels bring the maintained identity inventory to 904. These scopes are kept distinct because proposing a structure does not provide its electrochemistry, route evidence or polymer behavior.

The experimental panel also samples a different chemical distribution. Twenty-two of its 39 endpoints are quinones; nine imide endpoints come from six molecules in one study, including PMDI and phthalimide derivatives. Viologen is represented by one molecule and NDI by none. Thus overall benchmark accuracy cannot be assigned to every enumerated family or exact graft. This imbalance is part of the applicability domain of the calculations, rather than a reason to omit less favorable endpoint errors.

3.2 The benchmark separates reference-like bias from step-dependent error

The unfitted protocol gives MAE 0.247 V, RMSE 0.279 V and bias +0.190 V (Figure 2). Molecule-cluster 95% bootstrap intervals are [0.213,0.286], [0.245,0.313] and [0.128,0.257] V, respectively. The seven unequal study clusters give less well calibrated summaries, reported separately in the SI. These are descriptive intervals for this retrospective panel.

All 30 first-step errors are positive, with mean +0.266 V, whereas the nine second-step errors have mean −0.064 V and MAE 0.185 V. A common offset therefore cannot explain the complete residual structure. In particular, all nine paired step spacings are too large in magnitude, and the Fc contribution cancels from each spacing. The apparent improvement of second-step MAE can include error cancellation rather than better second-step prediction. Reference, solvation and charge-state contributions must be assessed together.

Post hoc molecule-held-out constant offsets reduce MAE/RMSE to 0.158/0.211 V; separate first-/second-step offsets give 0.106/0.166 V. Study-held-out values are 0.182/0.232 and 0.124/0.177 V. These internal diagnostics establish systematic residual structure but are not adopted as prospective corrections. The computed Fc absolute value is 4.379 V. Reference-only substitutions from other functionals are not level-matched alternatives, and fixed-geometry B3LYP/PBE0 controls worsen the counterfactual benchmark bias (SI). Related work in DMF likewise found different first-/second-reduction intercepts under a different protocol and reference.[11]

The five neutral-partner diffuse-basis controls shift first potentials by 11.7–35.1 mV when anion and reference records are fixed. Rotor-cutoff variation gives a maximum endpoint span of 25.4 mV and aggregate MAEs of 0.245–0.247 V. These measured sensitivities are substantially smaller than the panel's overall MAE, while addressing only specific protocol choices. Neither establishes basis convergence, a unique source of the residual bias or a candidate-specific prediction interval.

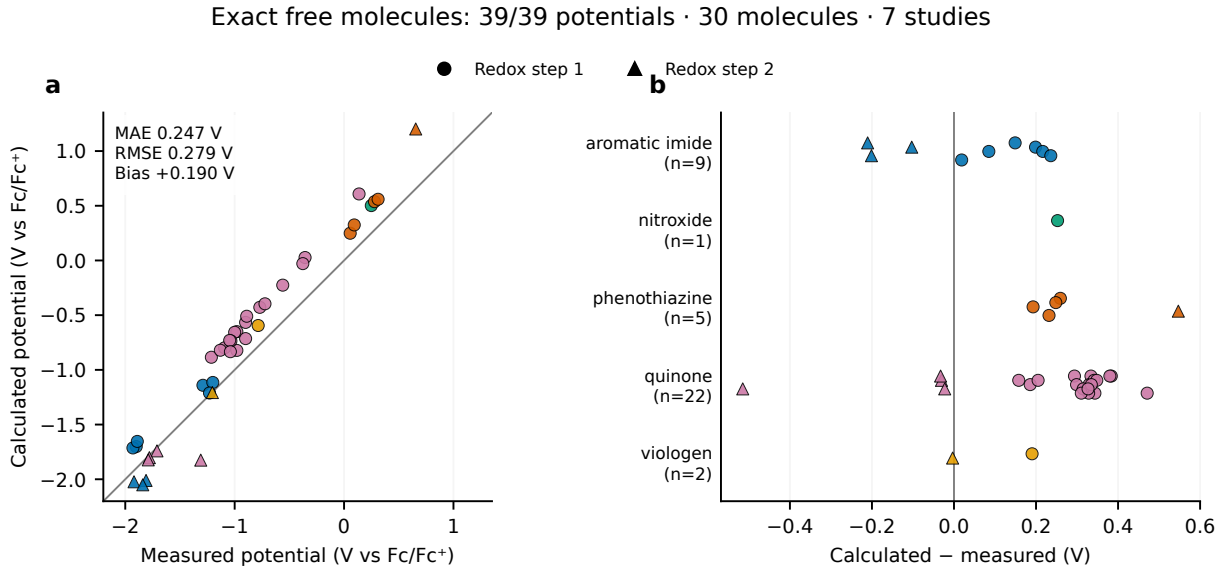


Figure 2: Retrospective benchmark for 39 sourced free-parent potentials in MeCN. All endpoints and outliers are retained, and no fitted offset is applied. Family and redox-step labels distinguish the panel's uneven coverage. Accuracy on these free parents does not validate the exact grafts or oligomer environments.

3.3 Route-admitted examples expose different missing dimensions

The baseline N-methyl viologen graft is exactly the enumerated design `search_viologen_b7a4dca7a5e1`. Kim and co-workers report preparation of its mono-methyl bipyridinium precursor and attachment to preformed Merrifield resin in refluxing MeCN, followed by separate counterion exchange.[3] This is evidence for the support-level attachment class. The computed 4-methylbenzyl product is a local model of that connection, and the reported approximate conversion is not a precisely measured loading for our material.

The checked molecular ladder contains the dicationic singlet, radical-cation doublet and neutral singlet. The referenced reductions are −0.608 and −1.208 V, separated by 0.600 V. The corresponding balanced disproportionation value is 57.91 kJ mol^{-1}. These values describe thermodynamic charge-state bookkeeping; they do not establish resistance to chemical decomposition or operational reversibility.

The three checked mono-methyl precursor seeds close the separated-ion quaternization cycle at −11.49 kJ mol^{-1} at the primary cutoff, with a 1.50 kJ mol^{-1} precursor-seed span. Cutoff settings give −13.30 to −11.49 kJ mol^{-1}. The calculation avoids an HCl/base-capture term because it represents quaternization, not neutral N–H/O–H substitution. Its favorable local-model value remains conditional on separated spectator ions, omitted counterion association and the modeled 298.15 K temperature; the synthesis precedent uses reflux. These differences preclude interpreting the value as a grafting yield.

The O-linked TEMPO graft provides an oxidation-active comparison. Its source-supported Merrifield attachment retains the nitroxide and uses NaH in DMF or THF.[36] The exact local-model oxidation potential is +0.579 V versus Fc/Fc$^+$. Independently checked directional solution-

relaxation costs are 1.477 eV for reduction and 1.639 eV for oxidation. Their 10.4% asymmetry exceeds the declared 5% equal-curvature diagnostic, so a single symmetric parameter remains unqualified. This is useful candidate information: successful electronic calculations do not justify discarding directional response or treating it as a measured transfer rate.

Table 1: Complementary exact-graft cases and the evidence they supply. All potentials refer to the computed Fc couple in MeCN. Reaction energies retain their stated finite-model convention.

Case	Checked molecular/reaction result	Remaining decision-relevant evidence
Viologen	Two reductions; local quaternization -11.49 kJ mol^{-1}	Ion association, polymer free energies, accessibility and transport
TEMPO	Oxidation; two directional solution-relaxation costs	Actual-condition grafting, polymer occupancy and transfer kinetics
PMDI-ethyl	Two reductions; finite first-site and neighboring-site cycles	Exact graft selectivity, broader sampling and material utilization

3.4 Reaction definition controls absolute grafting signs

Of the 75 historical local reference reactions, 67 have complete free energies and eight remain held. Five complete candidate reactions and six checked DBU/DIPEA components give twenty base-assisted values. The separated-ion capture terms are -102.78 kJ mol^{-1} for DBU and -70.98 kJ mol^{-1} for DIPEA. Under Eq. 7, each base representation shifts every matched reaction by the same term; changing that term cannot reorder candidates within the representation.

The two highlighted benzoquinone HCl-reference values are $+57.63$ and $+56.03$ kJ mol^{-1}. They become $-45.14/-46.75$ kJ mol^{-1} with DBU and $-13.34/-14.94$ kJ mol^{-1} with DIPEA (Figure 3). AQ-1-oxy and NDI also change sign in the panel. The relevant inference is that an HCl-reference sign cannot exclude an unmodeled base-assisted synthesis, while a favorable base-assisted model cannot supply missing precursor or selectivity evidence.

External comparisons qualify the absolute values. The revised MeCN DBU/basicity and recommended HCl scales imply a capture term of -79.97 kJ mol^{-1}, differing from the calculation by -22.81 kJ mol^{-1}.[30–32] In contrast, HCl/chloride-free relative-basicity controls for pyridine and triethylamine against DBU differ by $+4.58$ and $+1.40$ kJ mol^{-1} from the corresponding tabulated values. A DIPEA spectrophotometric measurement gives $\log K^{\mathrm{H}} = 18.64(1)$ in MeCN at 25 °C; a separate NMR salt estimate gives 18.7 in MeCN-d_3.[33, 34] Combining the former with the DBU scale gives $+32.36$ kJ mol^{-1} for relative proton transfer versus $+31.80$ kJ mol^{-1} computed, whereas its HCl-capture comparison gives -47.60 versus -70.98 kJ mol^{-1}. Indicator, activity and association conventions are not established to match across sources.

These controls show why agreement for one relative reaction does not validate every absolute grafting sum. Replacing only the DBU capture term gives $-22.33/-23.94$ kJ mol^{-1} for the two BQ reactions, but changes the cancellation of HCl-related errors in the complete cycle. Such hybrid values remain sensitivities. All 87 complete historical reference/base-assisted values retain their nominal sign over the three rotor cutoffs, with a maximum span of 4.46 kJ mol^{-1}; that numerical robustness does not close the independent speciation uncertainty.

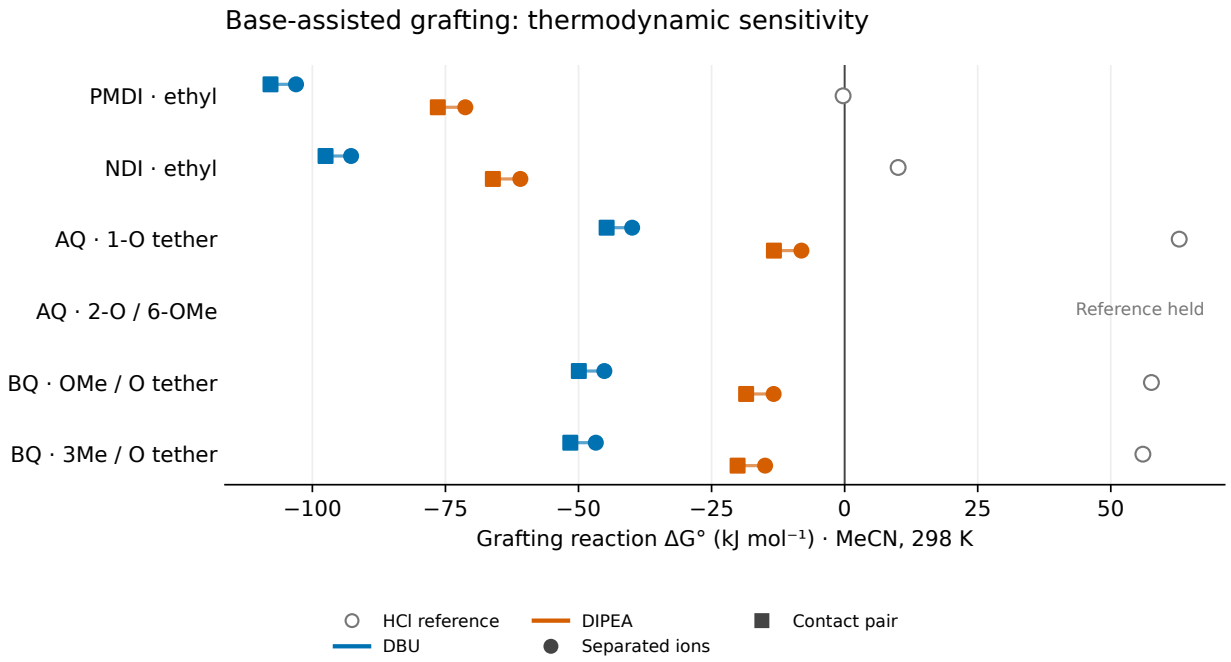


Figure 3: Local substitution energies under HCl-reference, separated-ion and contact-pair conventions. Each specified base contributes a shared term to compatible reactions. Changing absolute signs does not establish a new within-base ranking, measured ion-pair population or experimental grafting yield.

3.5 PMDI distinguishes a first-site polymer penalty from a neighboring-graft response

The PMDI-ethyl local graft has checked reductions at -1.076 and -1.977 V versus Fc. Its HCl-reference attachment value is -0.29 kJ mol^{-1}, and the defined DBU/separated-ion model gives -103.07 kJ mol^{-1}. The exact mono-N-ethyl/NH precursor is reported in Example 19 of EP0010440B1, and molecular PMDI N-alkylation has separate precedent.[37, 38] These sources support access and reaction-class plausibility, but not selective DBU-assisted installation on the exact bead or retention of its redox behavior. This case therefore tests linked computational comparisons without declaring a completed synthesis route.

For the sampled five-repeat minima, attachment at sites 2 and 3 gives -97.7 and -97.0 kJ mol^{-1} under the same DBU/separated-ion convention. Relative to the capped reaction, the polymer penalties are $+5.34$ and $+6.11$ kJ mol^{-1} (Eq. 8). Their positivity persists across rotor cutoffs, but the small relative preference between the two sites changes sign (Figure 4). Common precursor and base terms cancel from the polymer-minus-local comparison. Thus the calculation identifies a finite-model context penalty more defensibly than it identifies a preferred graft site.

The four-state occupancy cycle gives $\Delta\Delta G_{23} = -1.86$ kJ mol^{-1} at the primary cutoff. Attachment at site 3 changes from -96.96 kJ mol^{-1} with site 2 empty to -98.82 kJ mol^{-1} with site 2 grafted; reversing the order closes the same double-attachment sum. Values at the three cutoffs are -1.07, -1.86 and -1.82 kJ mol^{-1} (Figure 5). The shared reaction terms cancel, so this non-additivity is identical for the specified separated-ion and contact-pair bookkeeping. It remains a response of the entire sampled chain and solvent.

The paired minimum has PMDI core centroids 16.30 Å apart. Adjacent sequence sites therefore

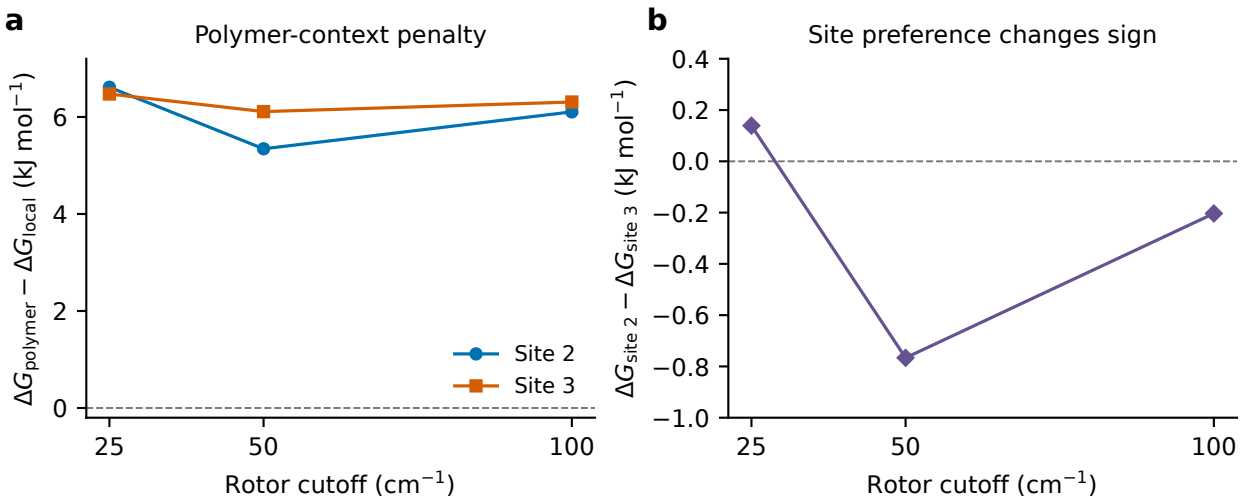


Figure 4: First-site PMDI attachment relative to the local capped reference. Polymer-minus-local penalties remain positive over tested rotor cutoffs, whereas the small difference between sites 2 and 3 changes sign. Lines connect numerical settings, not statistical uncertainty bounds. Each occupied graph contributes one minimum.

do not imply close-contact stacking, and the weak cycle response cannot be read as an isolated PMDI binding energy or coupling. Each of the four graphs contributes one minimum; broader occupied sampling is required before assigning equilibrium populations or a general spacing preference.

Polymer mass introduces another independent distinction. The neutral capped PMDI molecule has a nominal two-electron capacity of 153.9 mAh g^{-1}. With three handles per five-repeat fragment, one, two or three grafts give dry-fragment ceilings of 61.2, 98.9 and 124.5 mAh g^{-1} when scaffold and residual chloride mass are retained. These bounds assume full use of two electrons per motif and omit unspecified crosslinker, electrolyte and charging counterion mass. They do not infer an experimental loading distribution or accessible capacity. A molecular descriptor can therefore support a redox comparison while overestimating the capacity relevant to the whole material.

3.6 Sequence spacing does not determine spatial contact

The finite phthalimide control supplies two accepted poses for each of six occupancy graphs. Its twelve native free energies give adjacent and one-spacer nonadditivities of −5.17 and −11.39 kJ mol^{-1} in the declared two-pose partitions. Both spacings support close-core and separated-core minima (Figure 6). Adjacent-site centroids are 4.24 and 16.66 Å apart; one-spacer centroids are 4.77 and 20.20 Å apart. The close-contact poses have lower G by 11.9 and 7.1 kJ mol^{-1} within their respective graph-specific pairs.

The geometry comparison motivates a separate interaction calculation. At the central cap length, frozen methyl-capped fragments from the close adjacent and one-spacer poses give MeCN electronic interactions of −24.76 and −26.22 kJ mol^{-1}, whereas separated poses are near zero. Those electronic sums include caps and continuum response and lack counterpoise correction. They cannot be subtracted from the whole-chain free-energy cycle to assign a backbone contribution, nor used as thermal association energies or electronic couplings.

Adding the second known pose lowers each finite partition by at most 0.14 kJ mol^{-1}, but this bounds only the contribution of that known pose. It does not bound undiscovered conformers.

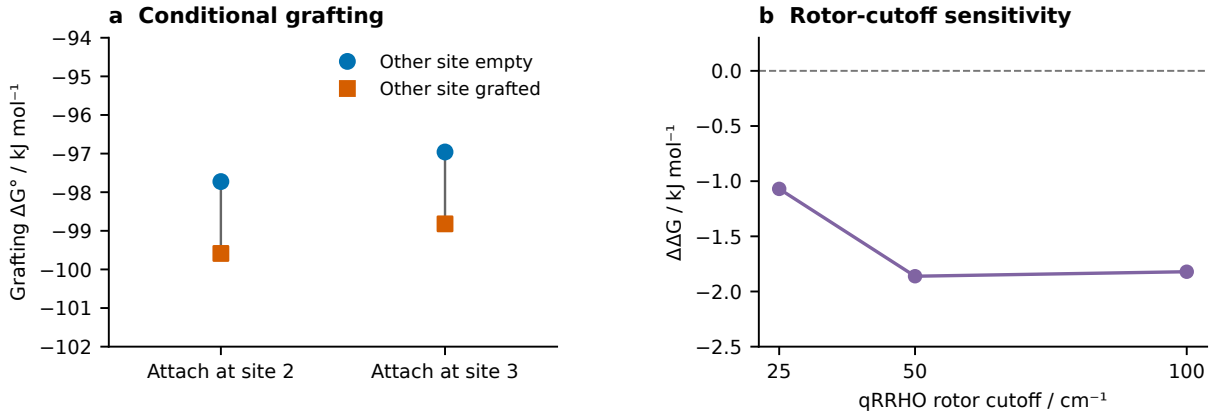


Figure 5: Matched PMDI neighboring-site cycle. Conditional attachment energies compare the same target site with the other site empty or grafted. Shared precursor, base and byproduct terms cancel from nonadditivity. The cutoff dependence reflects numerical scenarios; one minimum per graph and separated cores delimit the interpretation.

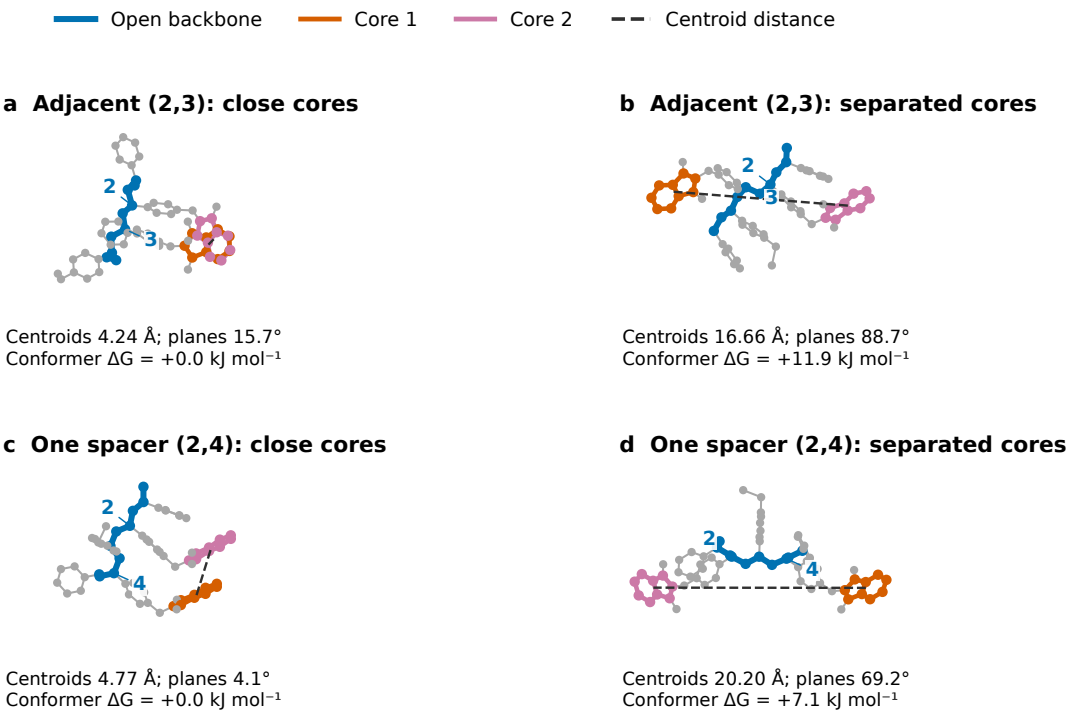


Figure 6: Actual phthalimide control conformers for adjacent and one-spacer occupancy. Backbone and motif colors identify the open chain and conjugated cores; unoccupied reactive sites retain benzyl chloride. Views use the saved coordinates, proper rotations and a common scale. Distances join core centroids, and relative free energies compare poses of the same chemical graph. These finite samples do not establish converged contact populations.

Rotor-cutoff changes give adjacent and one-spacer cycle ranges of −5.17 to −4.64 and −12.34 to −11.39 kJ mol^{-1}. Expanded sampling in the separate smooth-surface MeCN control further illustrates the component-wise requirement: its two empty-backbone sets meet the declared 10-to-20 input diagnostic, with changes of −1.03 and −1.23 kJ mol^{-1}, while occupied sets remain incomplete. This does not qualify occupied ensemble cycles or permit mixing the alternate-model values into the primary cycles.

3.7 Implications for candidate selection and method applicability

The candidate examples expose different limits on a seemingly favorable molecular design. Viologen combines a reviewed route with two checked reductions and a favorable local quaternization model, yet its ionic polymer environment remains to be qualified. TEMPO combines route evidence and a checked oxidation potential with directional electronic response that resists reduction to a symmetric parameter. PMDI provides explicit finite polymer cycles but retains a practical-route hold. A scalar score that fills these missing dimensions from family analogues would erase scientifically relevant differences among the candidates.

The resulting assessment requires every application-specific criterion to be evaluated in a declared reaction or operating context. Exact identity connects those contexts; it does not make reaction solvent and battery electrolyte identical. Operating redox window, accessible capacity, state stability, intended charge transfer, attachment feasibility, defined-route thermodynamics, unwanted association, coupling, occupancy sensitivity and accessibility remain distinct requirements. A failed criterion excludes the design in that context, whereas missing evidence or an undeclared threshold leaves the material decision pending. The historical 878-candidate all-axis ledger and the broader identity inventory describe different scopes, and neither currently supplies a complete material shortlist.

The principal transferable result is a disciplined comparison between chemical models. Base shifts should be interpreted under their shared stoichiometry; polymer penalties should use matched local references; neighboring-site responses should use balanced occupancy graphs; and contact geometry should be accompanied by an appropriate interaction descriptor. These requirements apply beyond the present motifs, although their practical utility must still be tested on additional chemistries and experimental grafts. Our study does not demonstrate improved intrinsic DFT accuracy over alternative functionals or superiority of a new screening algorithm.

The remaining uncertainties also define useful next calculations. Counterion and explicit-solvent sensitivity are particularly important for charged grafts and substitution salts. Conformer/configuration coverage must be completed separately for every component of an occupied cycle. Electronic coupling and transport require a declared charge-transfer mechanism rather than inference from neutral binding: polymer electrodes can exploit electron hopping, and high charge propagation has been demonstrated in TEMPO layers.[13, 39] Concentration and backbone effects observed in phthalimide polymers further caution against transferring a local molecular response directly to a material.[14] Experimental loading, retained redox activity and electrolyte accessibility are ultimately needed to connect these computational scenarios to synthesis priorities.

4 Conclusions

We have connected molecular electrochemistry, defined attachment reactions and finite polymer occupancy models through exact chemical identities and explicit calculation contexts. The benchmark establishes the present protocol's retrospective accuracy and its uneven chemical coverage. Viologen, TEMPO and PMDI then demonstrate complementary uses of the linked assessment:

route-specific quaternization, directional solution response, and separation of a first-site polymer penalty from a neighboring-graft free-energy response. Phthalimide controls show that a sequence spacer can coexist with close spatial contact, requiring geometry, electronic interaction and conditional thermodynamics to retain their separate meanings. The framework supports reproducible comparisons and identifies missing evidence without promoting a favorable descriptor into a completed polymer-material prediction.

Supporting Information

Benchmark sources, reference handling, bootstrap and offset diagnostics; basis and rotor sensitivity; exact candidate and route identities; native-state quality checks; solution-relaxation cycles; grafting components and occupancy partitions; fragment interactions; numerical implementation, restart and portable replay details (PDF).

Data and Software Availability

Versioned code, curated endpoint data and hash-verified evidence snapshots are available in the project repository. The exact-viologen CPU example supplies the ten-state geometries, electronic energies, Hessians and thermochemical records needed to reconstruct the reported molecular redox and local quaternization results using NumPy. Additional benchmark, grafting and oligomer records are assembled in source-bound reviewer packages described in the SI. The computational environments and source inventories are specified there. Full raw-data archival deposition, a permanent release identifier and final deposition metadata remain outstanding and must be completed before submission of a final version.

Author Information

Authorship, affiliations, corresponding-author details, contributions, funding and competing-interest statements require confirmation by the authors.

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OpenAI Codex was used during manuscript preparation to assist with literature retrieval, drafting and revision of prose, workflow code and numerical-check scripts. Computational claims are tied to the recorded calculations and independent checks described in the Methods and Supporting Information. The authors must review and approve the complete scientific content and this disclosure before submission.

References

[1] Tobias Janoschka, Norbert Martin, Udo Martin, Christian Friebe, Sabine Morgenstern, Hannes Hiller, Martin D. Hager, and Ulrich S. Schubert. An aqueous, polymer-based redox-flow battery using non-corrosive, safe, and low-cost materials. *Nature*, 527:78–81, 2015. doi: 10.1038/nature15746.

[2] Elena C. Montoto, Gavvalapalli Nagarjuna, Jingshu Hui, Mark Burgess, Nina M. Sekerak, Kenneth Hernández-Burgos, Teng-Sing Wei, Marissa Kneer, Joshua Grolman, Kevin J. Cheng, Jennifer A. Lewis, Jeffrey S. Moore, and Joaquín Rodríguez-López. Redox Active Colloids as Discrete Energy Storage Carriers. *Journal of the American Chemical Society*, 138:13230–13237, 2016. doi: 10.1021/jacs.6b06365.

[3] Dukhan Kim, Melanie S. Sanford, Thomas P. Vaid, and Anne J. McNeil. A nonaqueous redox-matched flow battery with charge storage in insoluble polymer beads. *Chemistry—A European Journal*, 28:e202200149, 2022. doi: 10.1002/chem.202200149.

[4] Algin Oh Biying, Sabrina Trano, Yi Jie Eng, Gabriele Lingua, Haritz Sardon, David Mecerreyes, and Julian M. W. Chan. Upcycling commodity styrofoam into redox polymer-based electrodes. *ChemSusChem*, 18(11):e202500092, 2025. doi: 10.1002/cssc.202500092.

[5] Süleyman Er, Changwon Suh, Michael P. Marshak, and Alán Aspuru-Guzik. Computational design of molecules for an all-quinone redox flow battery. *Chemical Science*, 6:885–893, 2015. doi: 10.1039/c4sc03030c.

[6] Rebekah Duke, Vinayak Bhat, Parker Sornberger, Susan A. Odom, and Chad Risko. Towards a comprehensive data infrastructure for redox-active organic molecules targeting non-aqueous redox flow batteries. *Digital Discovery*, 2:1152–1162, 2023. doi: 10.1039/d3dd00081h.

[7] Elif Sorkun, Qi Zhang, Abhishek Khetan, Murat Cihan Sorkun, and Süleyman Er. RedDB, a computational database of electroactive molecules for aqueous redox flow batteries. *Scientific Data*, 9:718, 2022. doi: 10.1038/s41597-022-01832-2.

[8] Andreas J. Achazi, Xhesilda Fataj, Philip Rohland, Martin D. Hager, Ulrich S. Schubert, and Doreen Mollenhauer. Development of a multi-step screening procedure for redox active molecules in organic radical polymer anodes and as redox flow anolytes. *Journal of Computational Chemistry*, 45(14):1112–1129, 2024. doi: 10.1002/jcc.27299.

[9] Aleksandr V. Marenich, Junming Ho, Michelle L. Coote, Christopher J. Cramer, and Donald G. Truhlar. Computational electrochemistry: prediction of liquid-phase reduction potentials. *Phys. Chem. Chem. Phys.*, 16:15068–15106, 2014. doi: 10.1039/c4cp01572j.

[10] Mansoor Namazian, Ching Yeh Lin, and Michelle L. Coote. Benchmark Calculations of Absolute Reduction Potential of Ferricinium/Ferrocene Couple in Nonaqueous Solutions. *Journal of Chemical Theory and Computation*, 6:2721–2725, 2010. doi: 10.1021/ct1003252.

[11] Sarah Elhajj and Samer Gozem. First and Second Reductions in an Aprotic Solvent: Comparing Computational and Experimental One-Electron Reduction Potentials for 345 Quinones. *Journal of Chemical Theory and Computation*, 20:6227–6240, 2024. doi: 10.1021/acs.jctc.4c00602.

[12] Cedrik Wiberg, Lars Evenäs, Michael Busch, and Elisabet Ahlberg. Naphthalene diimides (NDI) in highly stable ph-neutral aqueous organic redox flow batteries. *Journal of Electroanalytical Chemistry*, 896:115224, 2021. doi: 10.1016/j.jelechem.2021.115224.

[13] Abhinav Akhoury, Lev Bromberg, and T. Alan Hatton. Interplay of Electron Hopping and Bounded Diffusion during Charge Transport in Redox Polymer Electrodes. *The Journal of Physical Chemistry B*, 117:333–342, 2013. doi: 10.1021/jp302157g.

[14] Khirabdhi Tannaya Mohanty, Sheila Keating, Kha Trinh, Riccardo Alessandri, Daniel P. Tabor, Stuart J. Rowan, Juan J. de Pablo, and Jodie L. Lutkenhaus. Deciphering concentration and backbone identity effects on the redox properties of phthalimide-based polymers. *Chemistry of Materials*, 38(18):9530–9541, 2026. doi: 10.1021/acs.chemmater.6c01965.

[15] Sebastiaan P. Huber, Spyros Zoupanos, Martin Uhrin, Leopold Talirz, Leonid Kahle, Rico Häuselmann, Dominik Gresch, Tiziano Müller, Aliaksandr V. Yakutovich, Casper W. Andersen, Francisco F. Ramirez, Carl S. Adorf, Fernando Gargiulo, Snehal Kumbhar, Elsa Passaro, Conrad Johnston, Andrius Merkys, Andrea Cepellotti, Nicolas Mounet, Nicola Marzari, Boris Kozinsky, and Giovanni Pizzi. AiiDA 1.0, a scalable computational infrastructure for automated reproducible workflows and data provenance. *Scientific Data*, 7:300, 2020. doi: 10.1038/s41597-020-00638-4.

[16] Anubhav Jain, Shyue Ping Ong, Wei Chen, Bharat Medasani, Xiaohui Qu, Michael Kocher, Miriam Brafman, Guido Petretto, Gian-Marco Rignanese, Geoffroy Hautier, Daniel Gunter, and Kristin A. Persson. FireWorks: a dynamic workflow system designed for high-throughput applications. *Concurrency and Computation: Practice and Experience*, 27:5037–5059, 2015. doi: 10.1002/cpe.3505.

[17] Daniel G. A. Smith, Doaa Altarawy, Lori A. Burns, Matthew Welborn, Levi N. Naden, Logan Ward, Sam Ellis, Benjamin P. Pritchard, and T. Daniel Crawford. The MolSSI QCA rchive project: An open-source platform to compute, organize, and share quantum chemistry data. *WIREs Computational Molecular Science*, 11:e1491, 2021. doi: 10.1002/wcms.1491.

[18] RDKit developers. RDKit: Open-Source Cheminformatics Software. https://www.rdkit.org/. Software; version used for current graph processing: 2026.3.5. Accessed 26 October 2026.

[19] James W. Furness, Aaron D. Kaplan, Jinliang Ning, John P. Perdew, and Jianwei Sun. Accurate and numerically efficient r2SCAN meta-generalized gradient approximation. *The Journal of Physical Chemistry Letters*, 11:8208–8215, 2020. doi: 10.1021/acs.jpclett.0c02405.

[20] Eike Caldeweyher, Sebastian Ehlert, Andreas Hansen, Hagen Neugebauer, Sebastian Spicher, Christoph Bannwarth, and Stefan Grimme. A generally applicable atomic-charge dependent London dispersion correction. *The Journal of Chemical Physics*, 150:154122, 2019. doi: 10.1063/1.5090222.

[21] Aleksandr V. Marenich, Christopher J. Cramer, and Donald G. Truhlar. Universal solvation model based on solute electron density and on a continuum model of the solvent defined by the bulk dielectric constant and atomic surface tensions. *The Journal of Physical Chemistry B*, 113:6378–6396, 2009. doi: 10.1021/jp810292n.

[22] Narbe Mardirossian and Martin Head-Gordon. ωB97M-V: A combinatorially optimized, range-separated hybrid, meta-GGA density functional with VV10 nonlocal correlation. *The Journal of Chemical Physics*, 144:214110, 2016. doi: 10.1063/1.4952647.

[23] Florian Weigend and Reinhart Ahlrichs. Balanced basis sets of split valence, triple zeta valence and quadruple zeta valence quality for H to Rn: Design and assessment of accuracy. *Physical Chemistry Chemical Physics*, 7:3297–3305, 2005. doi: 10.1039/B508541A.

[24] Dmitrij Rappoport and Filipp Furche. Property-optimized Gaussian basis sets for molecular response calculations. *The Journal of Chemical Physics*, 133:134105, 2010. doi: 10.1063/1.3484283.

[25] Qiming Sun et al. Recent developments in the PySCF program package. *The Journal of Chemical Physics*, 153:024109, 2020. doi: 10.1063/5.0006074.

[26] Rui Li, Qiming Sun, Xing Zhang, and Garnet Kin-Lic Chan. Introducing GPU-acceleration into the Python-based simulations of chemistry framework, 2024. Preprint.

[27] Lee-Ping Wang and Chenchen Song. Geometry optimization made simple with translation and rotation coordinates. *The Journal of Chemical Physics*, 144:214108, 2016. doi: 10.1063/1.4952956.

[28] Stefan Grimme. Supramolecular Binding Thermodynamics by Dispersion-Corrected Density Functional Theory. *Chemistry – A European Journal*, 18:9955–9964, 2012. doi: 10.1002/chem.201200497.

[29] Christoph Bannwarth, Eike Caldeweyher, Sebastian Ehlert, Andreas Hansen, Philipp Pracht, Jakob Seibert, Sebastian Spicher, and Stefan Grimme. Extended tight-binding quantum chemistry methods. *WIREs Computational Molecular Science*, 11:e1493, 2021. doi: 10.1002/wcms.1493.

[30] Sofja Tshepelevitsh, Agnes Kütt, Märt Lõkov, Ivari Kaljurand, Jaan Saame, Agnes Heering, Paul G. Plieger, Robert Vianello, and Ivo Leito. On the basicity of organic bases in different media. *European Journal of Organic Chemistry*, 2019:6735–6748, 2019. doi: 10.1002/ejoc.201900956.

[31] Elin Raamat, Karl Kaupmees, Gea Ovsjannikov, Aleksander Trummal, Agnes Kütt, Jaan Saame, Ivar Koppel, Ivari Kaljurand, Lauri Lipping, Toomas Rodima, Viljar Pihl, Ilmar A. Koppel, and Ivo Leito. Acidities of strong neutral Brønsted acids in different media. *Journal of Physical Organic Chemistry*, 26:162–170, 2013. doi: 10.1002/poc.2946.

[32] University of Tartu, Chair of Analytical Chemistry. Author-maintained basicity collection and recommended acid table. https://analytical.chem.ut.ee/databases/pka-values/. Accessed 22 October 2026. Acid table: https://analytical.chem.ut.ee/HA_UT/pKa_Table_Water_Acetonitrile_DCE_Gas_Phase.pdf.

[33] D. Barišić, N. Cindro, N. Vidović, N. Bregović, and V. Tomišić. Protonation and anion-binding properties of aromatic sulfonylurea derivatives. *RSC Advances*, 11(39):23992–24000, 2021. doi: 10.1039/D1RA04738H.

[34] Emma N. Cook, Ian M. Courter, Diane A. Dickie, and Charles W. Machan. Controlling product selectivity during dioxygen reduction with Mn complexes using pendent proton donor relays and added base. *Chemical Science*, 15(12):4478–4488, 2024. doi: 10.1039/D3SC02611F.

[35] Zhi-Qiang You, Jan-Michael Mewes, Andreas Dreuw, and John M. Herbert. Comparison of the marcus and pekar partitions in the context of non-equilibrium, polarizable-continuum solvation models. *The Journal of Chemical Physics*, 143(20):204104, 2015. doi: 10.1063/1.4936357.

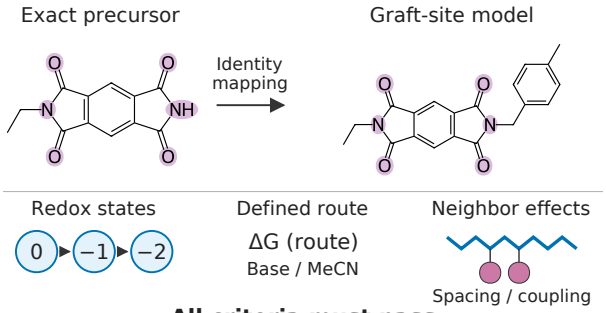
[36] Reinhard Sommerlade, Hansjörg Grützmacher, and Souâd Boulmaâz. Process for the selective oxidation of alcohols using readily removable nitroxyl radicals. U.S. Patent 6,441,243 B1, Example A4, 2002. URL https://patents.google.com/patent/US6441243B1/en.

[37] Richard J. Bochis, Michael H. Fisher, and Bruce O. Linn. Pyromellitic diimides, processes for their preparation, these compounds in orally administrable form for use in increasing the feed efficiency of a ruminant animal and compositions for this purpose containing them. European patent EP0010440B1, published January 26, 1983, 1983. URL https://patents.google.com/patent/EP0010440B1/en. Example 19, printed p. 10; precursor identity only.

[38] Darren G. Hamilton, Daniel E. Lynch, Karl A. Byriel, Colin H. L. Kennard, and Jeremy K. M. Sanders. N-alkylation of pyromellitic diimide: Solid-state structure of the 1 : 1 N,N′-diethylpyromellitic diimide/2,6-dimethoxynaphthalene cocrystal. *Australian Journal of Chemistry*, 51(5):441–444, 1998. doi: 10.1071/C97214.

[39] Kenichi Oyaizu, Yuko Ando, Hiroaki Konishi, and Hiroyuki Nishide. Nernstian Adsorbate-like Bulk Layer of Organic Radical Polymers for High-Density Charge Storage Purposes. *Journal of the American Chemical Society*, 130:14459–14461, 2008. doi: 10.1021/ja803742b.

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